VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (Matter_2016_Science_Carbfix)
==========

Title: Rapid carbon mineralization for permanent disposal of anthropogenic carbon dioxide emissions

Authors: Juerg M. Matter, Martin Stute, Sandra Ó. Snæbjörnsdóttir, Eric H. Oelkers, Sigurdur R. Gislason, et al. (2016)

Source: Science, Vol. 352, Issue 6291, pp. 1312–1314; & Nature Rev Earth Env 1, 90–102 (2020)

DOI/URL: 10.1126/science.aad8132 / 10.1038/s43017-019-0011-8

Sample: N = 230 tonnes CO2 injected (175 t pure CO2 + 73 t CO2-H2S mix); HN-02 injection, HN-04 monitoring

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

- In-situ mineralization kinetics: >95% of injected CO2 was mineralized into solid carbonate minerals (calcite CaCO3) within <2 years (<24 months) at depth 400-800 m and temperatures 20-50 °C.
- Dissolution trapping at wellhead: Pure CO2 dissolved in water at wellhead pressure P > 25 bar (hydrostatic 40-80 bar at depth 400-800 m), preventing free gas bubbling and buoyancy migration.
- Water co-injection ratio: Required ~25 tonnes of water (or seawater) per tonne of CO2 to ensure 100% aqueous dissolution of CO2(aq) at in-situ reservoir conditions.
- Geochemical reactions & thermodynamics: Basalt divalent cations (Ca2+, Mg2+, Fe2+) drive spontaneous carbonation: CaSiO3 (wollastonite) + CO2 -> CaCO3 (calcite) + SiO2 [Delta G° = -64.4 kJ/mol]; Mg2SiO4 (forsterite) + 2CO2 -> 2MgCO3 (magnesite) + SiO2 [Delta G° = -112.8 kJ/mol].
- IPCC SRCCS trapping hierarchy (Fig SPM.3 / 5.9): 1. Structural trapping (caprock, t=0); 2. Residual trapping (capillary pore throats, 1-50 yr); 3. Solubility trapping (brine dissolution, density increases 1-2%, 10-1000 yr); 4. Mineral trapping (calcite precipitation, permanent >10,000 yr).
- Geological contrast: Reactive basalts yield rapid mineralization in years, whereas quartz sandstone sedimentary basins lack divalent cations and require centuries to millennia.

VERBATIM SOURCE QUOTE / TABLE RECORD:

Matter et al. (Science 2016, p. 1312): 'We find that over 95% of the CO2 injected into the CarbFix site in Iceland was mineralized to carbonate minerals in less than 2 years. This result contrasts with the common view that the immobilization of CO2 as carbonate minerals within geologic reservoirs takes several hundreds to thousands of years... safe long-term storage of anthropogenic CO2 emissions through mineralization can be far faster than previously postulated.' Snæbjörnsdóttir et al. (Nat Rev Earth Env 2020, p. 90): '...injection into reactive rocks (such as mafic or ultramafic lithologies), provoking CO2 mineralization and, thereby, permanently fixing carbon with negligible risk of return to the atmosphere.'